

Inter-City Freight Load Matching Marketplace

Problem Statement #28 — Requirements Engineering & UML Use-Case Modelling

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a verified Shipper to post a freight shipment by entering cargo details, pickup location, delivery location, required vehicle type, pickup deadline, and auction timeframe.	High	Pass: Valid shipment is posted and visible to eligible carriers. Fail: Rejected if mandatory information is missing or shipper is not verified.	Allows shippers to advertise available freight loads and initiate matching.
FR-002	Functional	The system shall allow verified Freight Carriers with valid permits to view eligible freight shipments and submit competitive bids within the specified auction timeframe.	High	Pass: Valid bid is recorded and shipper notified. Fail: Unverified carrier or carrier with invalid permits cannot bid.	Enables competitive matching while ensuring only qualified carriers participate.
FR-003	Functional	The system shall allow the Shipper to review submitted bids and select a winning Freight Carrier before the auction deadline.	High	Pass: Selected carrier is notified and shipment becomes awarded. Fail: Selection is rejected after auction closes.	Gives the shipper control over carrier selection.
FR-004	Functional	The system shall create a shipment transaction and track predefined delivery milestones between the Shipper and selected Freight Carrier.	High	Pass: Pickup, transit, delivery, and proof-of-delivery milestones are recorded with timestamps. Fail: Unauthorized users cannot modify milestone records.	Provides shipment visibility and a reliable transaction history.
FR-005	Functional	The system shall automatically release the milestone payment held in escrow after the Freight Carrier submits valid electronic proof of delivery and delivery is verified.	High	Pass: Payment is released after valid proof is verified. Fail: Payment remains in escrow when verification fails.	Provides secure automated settlement and prevents premature payment.
NFR-001	Performance & Security	Milestone payments held in escrow shall be disbursed automatically within 60 seconds of verified electronic proof-of-delivery submission, while payment and shipment data shall be protected against unauthorized access.	High	Pass: Testing confirms release within 60 seconds under simulated peak load and prevents unauthorized payment access. Fail: Target latency/security is not met.	Fast and secure settlement is essential for a reliable B2B marketplace.
NFR-002	Availability & Reliability	The system shall maintain reliable availability during active shipment auctions and preserve accepted bids, shipment records, milestone information, and payment transactions without data loss.	High	Pass: Availability/recovery tests show active transactions remain accessible and data is preserved after simulated failures. Fail: Data is lost or unavailable.	Auctions and transactions are time-sensitive; downtime/data loss can cause operational and financial loss.

Use-Case Flow Specification

UC-002 — Submit Competitive Bid

Primary Actor: Freight Carrier

Supporting Actor: Shipper

Goal: Allow a verified carrier to submit a competitive bid for an available freight shipment during the active auction timeframe.

Preconditions

- The Freight Carrier has an account on the marketplace.
- The Freight Carrier is verified and has valid permits.
- A Shipper has posted an available freight shipment.
- The shipment auction period is active.
- The carrier has access to the marketplace.

Postconditions

- The bid is stored and associated with the shipment.
- The Shipper is notified of the new bid.
- The bid remains available for review until the auction closes.
- If submission fails, the bid is not recorded and the carrier is informed of the reason.

Main Success Scenario

1. The Freight Carrier logs into the marketplace.
2. The system authenticates the carrier.
3. The system verifies the carrier's verification status and permits.
4. The carrier browses available freight shipments.
5. The carrier selects a suitable freight shipment.
6. The system displays shipment details, including cargo, pickup location, delivery location, vehicle requirements, and auction deadline.
7. The carrier enters the proposed bid amount and required bid information.
8. The carrier submits the bid.
9. The system verifies that the auction timeframe is still active.
10. The system records the bid against the shipment.
11. The system confirms successful bid submission to the carrier.
12. The system notifies the Shipper about the newly submitted bid.
13. The Shipper can review the bid with other submitted bids.

Alternate Flow — AF-01: Auction Has Closed

1. At Step 9, the system determines that the auction deadline has passed.
2. The system rejects the bid submission.
3. The system informs the carrier that the auction is closed.
4. The bid is not recorded.

5. The carrier may return to the available shipment list and select another active shipment.

UML Use-Case Diagram — Inter-City Freight Load Matching Marketplace

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